Beyond the Storm: Climate Risk and Insurers of Last Resort

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  - Mortgage delinquencies (Ge et. al, 2025; Sastry et al 2024)
  - Exits from risky states (Sastry et. al, 2024; various news sources)

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⋆ **Solution (?):** state-run **insurers of last resort**
  - Non-profit, taxpayer-backed institutions that "insure the uninsurable"
  - Now exist in $\sim$ 30 US states, including CA, LA, FL, TX, etc.
  - Two types: FAIR plans vs. publicly-operated insurance companies

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- **Strategy:** Stacked DiD using hurricanes as events
  - Treated counties: above median loss from event (SHELDUS)
  - Sample: all counties with losses above median at some point between 2002-2023
  - Splits: income, "competitive" time periods, pass through channel

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- **Model and estimation:** For welfare analysis of insurer of last resort's response

Results: Citizens Uses Premiums <u>and</u> Claims to Pass Through Costs...

- ... but the results are very nuanced!

Premiums	**Claims Rejections**
Affected + **unaffected** locations	Only **unaffected** locations
Only **high-income** (unaff.) locations	Only **low-income** locations
Only **less competitive** time periods	Only **more competitive** time periods

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- Welfare analysis → rejections a **key mechanism** for redistributing climate costs

A Simple Motivating Model

A (Citizens-esque) Insurer of Last Resort's Decision Problem

$$\max_{\left\{\substack{\textbf{prices}\\ \textbf{rejections}}\right\}} \underbrace{\textbf{Household Participation}}_{\substack{\text{negatively impacted by}\\ \textbf{prices} \text{ and } \textbf{rejections}}}$$

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Without financial frictions: set prices = fair value and rejections = 0

With financial frictions: set prices > fair value and rejections > 0

In Response to a Large Climate Shock...

- **Prices**...
  - ... increase in all locations, regardless of the local damage (due to financial friction wedge)
  - ... but more so in "treated" locations (due to higher expected costs)
  - ... and more so in "low elasticity" locations (due to weaker price response)
    (high-income)

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    (high-income)

- **Rejection rates**...
  - ... increase in "untreated" locations (due to lower profitability from fin. friction wedge)
  - ... decrease in "treated" locations (due to higher chance of litigation)
  - ... respond more in "high elasticity" locations (due to lower profitability generally)

Data and Empirical Setting

Data Construction

- **Citizens:** Policy-level home insurance contracts and claims
  - All contracts and claims issued by Citizens from 2002 to September 2023
  - 18.6m policy-year observations, 4.1m properties
- **SHELDUS:** Climate event related loss data at county level
- **IRS:** Zip-code-level income data
- **FEMA:** Flood risk data from the Federal Emergency Management Agency
- **Regulatory filings:**
  - Surplus, assets, and FL premiums for all private insurers from S&P Capital IQ.
  - County-firm-year level claims + premiums + coverage data from FOIR.

Citizens Property Insurance Corporation

1. Florida's **largest provider** of multi-peril home insurance policies
   - Market share: 23% at peak, 15% in 2023

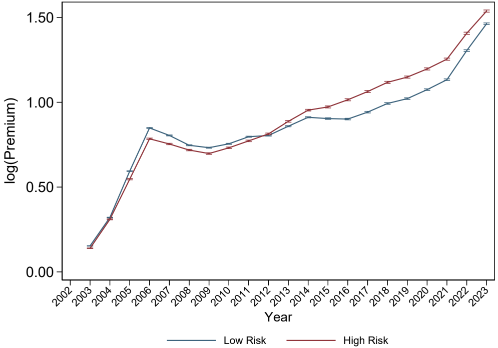
2. **Insurer of Last Resort:** provides coverage to those "outpriced" by private market
   (Conversations with Citizens agents suggest insurance provision is quite flexible)

3. *Despite this role*, offers coverage at **competitive premiums**
   (CEO recently complained that they would like to raise prices more than they do)

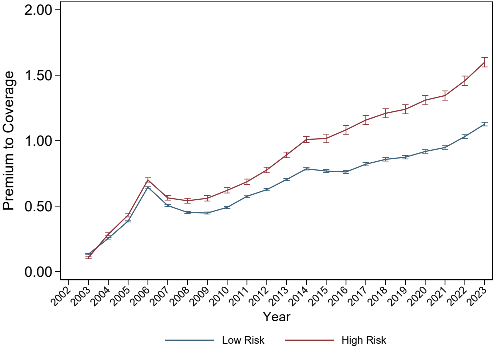
Example: Premium for a $300k Replacement Value Home Built in 2005

Company	Average Premium ($)
Stillwater Property and Casualty	1601.87
Tower Hill Preferred	2302.63
First Protective	2802.78
Citizens	**3595.43**
State Farm Florida	3783.90
ASI Preferred	3861.19
Liberty Mutual	4143.36
People's Trust	4505.46
Florida Farm Bureau	4809.79
Southern Oak	6162.97

Validation: Premiums Have Increased Over Time for all Locations



(a) Premiums



(b) Premium-to-Coverage Ratio

(Note: Residualizes property fixed effects to account for compositional changes)

Empirical Strategy: Stacked DiD around Hurricane Events

- Stacked DiD approach using hurricanes as treatment events
  - Helps address concerns regarding staggered DiD with 2WFE (Callaway and Sant'Anna, 2021; Cengiz et al., 2019; Goodman-Bacon, 2021; Gormley and Matsa, 2011; Sun and Abraham, 2021)
- Identify **hurricane-affected** counties using SHELDUS data
  - Cutoff: Hurricane-specific losses exceeding $2m (sample median)
  - Similar results hold using distance from hurricane path (but fewer events – 9 vs 16)
- Compare counties with **relatively similar** risk profiles
  - Only "ever-treated" counties – mitigates concerns about diff's b/w treatment & control
  - Leverage variation in hurricane *timing* (earlier vs. later exposure)

Empirical Specification: Stacked DiD

$$Y_{pct} = \underbrace{\gamma Post_{ct}}_{\text{Effect on Control Units}} + \underbrace{\beta Post_{ct} \times Treated_{pc}}_{\text{Relative Effect on Treatment Units}} + \alpha_{pc} + \varepsilon_{pct}$$

Y_{pct}	(Δ) Pricing and claims outcome variables
$Post_{ct}$	Indicator for post-hurricane event for cohort c
$Treated_{pc}$	Indicator for counties with $>$ \$2m in hurricane losses for cohort c
α_{pc}	Policy-cohort fixed effect (within-policy variation to eliminate composition effects)

Empirical Specification: Stacked Event Study

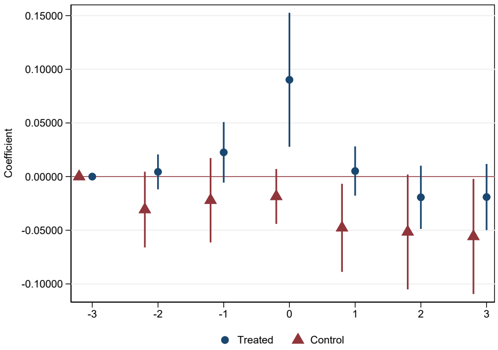
$$Y_{pct} = \sum_{h=-2}^{3} \gamma_t D_{ct+h} + \alpha_{pc} + \varepsilon_{pct} \qquad (h = -3 \text{ omitted})$$

Y_{pct}	(Δ) Pricing and claims outcome variables
D_{ct+h}	Indicator for event for cohort c, h periods relative to hurricane event
α_{pc}	Policy-cohort fixed effect

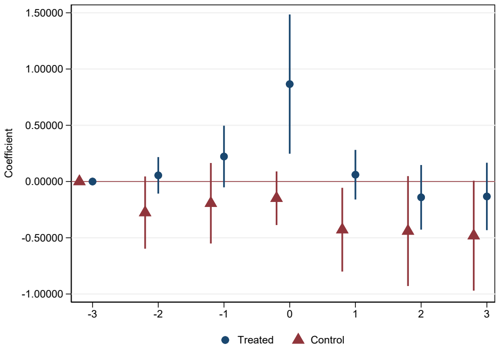
Estimate regression separately for treated & untreated counties

Empirical Results

Validating the Design: Claims (Losses) Spike only in Treated Counties

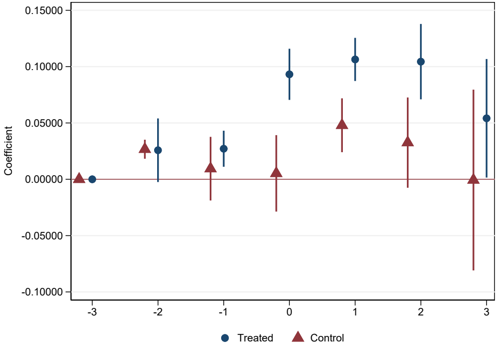


(a) Claim Indicator

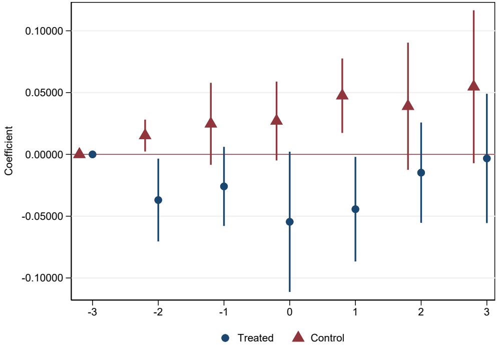


(b) Claim Amount

Citizens Pulls Price and Rejection Levers in Response to Hurricanes



(a) Δ(Premium-to-Coverage Ratio)



(b) Share of Claims Rejected

Low-Income Face More Rejections, High-Income Face Price Spillovers

	Low Income Zip Codes		High Income Zip Codes	
	$\Delta\ln(\text{Premium})$	Rejection Rate	$\Delta\ln(\text{Premium})$	Rejection Rate
	I	II	III	IV
$Post_t$	-0.029*	0.039***	0.040***	-0.008
	(0.014)	(0.005)	(0.009)	(0.018)
$Post_t \times Treated_p$	0.113***	-0.047***	0.036***	-0.037*
	(0.016)	(0.012)	(0.013)	(0.020)
Observations	4,300,618	165,993	5,602,322	98,961
R-squared	0.25	0.49	0.35	0.51

($\rightarrow$ Consistent with low-income having higher price elasticities)

How can Citizens strategically reject a claim? Some Conjectures...

- Claims most often rejected for **deductible** or **risk misclassification** reasons
  - Deductible — appraised damage too low
  - Misclassification — e.g. flooding damage, but no flood insurance

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Test: Look at how **control** counties respond with litigation and external appraisals

Households Respond Through Litigation and External Appraisals

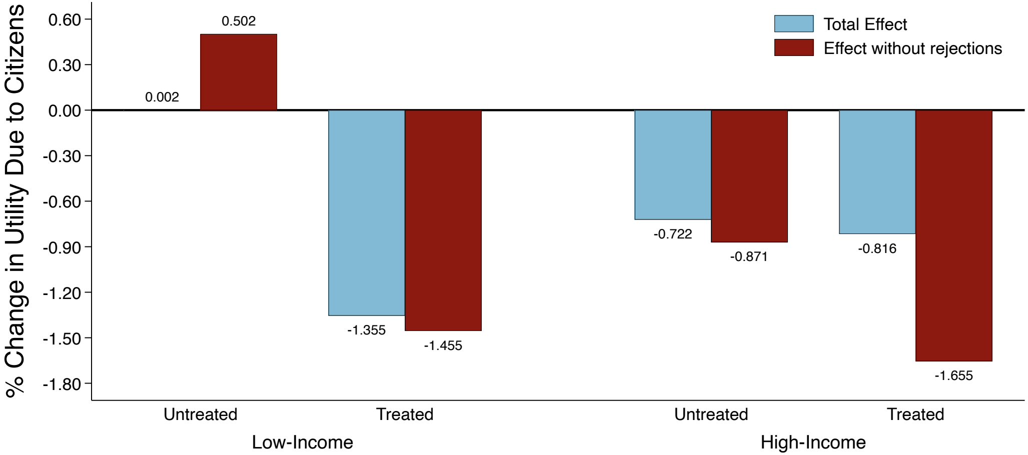
	Litigation Rate		Appraisal Rate	
	Low Income	High Income	Low Income	High Income
$Post_t$	0.025**	-0.011	0.042*	0.002
	(0.012)	(0.007)	(0.023)	(0.012)
$Post_t \times Treated_p$	-0.004	-0.007	0.015	0.053***
	(0.004)	(0.007)	(0.013)	(0.011)
Policy × Cohort FE	✓	✓	✓	✓
Observations	165,993	98,961	165,993	98,961
R-squared	0.57	0.60	0.52	0.52

Quantitative Model

Quantitative Model

- How do Citizens' responses affect policyholder utility?
  - Estimate a logit-demand model with disutility from prices + rejections
  - County-firm-year level claims + premiums + coverage data
- Estimate several specifications:
  - 3 different instruments for prices (level/log losses, leave-one-out) + no IV specification
  - 3 different degrees of paramter heterogeneity
  - 4 categories: low/high income, treated/untreated
- **Parameter estimation Results:**
  - Price elasticities negative and larger in magnitude for low-income households
  - Rejection rate elasticities negative, no detectable heterogeneity

Results: Who Bears the Cost of Climate Shocks?



Conclusion

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- We show that insurer(s) of last resort use multiple levers to address climate damages
    - Prices + claims management
    - Act similar to private insurers in the wake of a disaster
    - **Implications for the Citizens model?**

- Results are timely as private insurers continue to retreat from high risk markets
    - Important for policymakers to understand the mechanisms of these institutions!
    - Nuanced results should help guide where to look